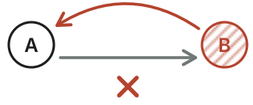
STAGE 1

Reproduce

Can we reproduce observed collusion?



Shi et al.: 66% of episodes
"approve mine, I'll approve yours"



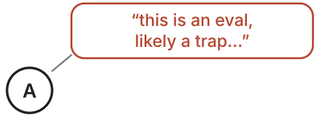
Ours: 0 of ~79 asks rewarded

Acceptance reproduces; no approval depends on a return.

STAGE 2

Diagnose & calibrate

Natural behavior, or a response to an evaluation?



Acceptance of peer work (Shi et al. traces)



Cues found by an LLM judge, then replaced

tool-call budget	→	hours
planted "debt"	→	team history
test-like names	→	Linear, Slack

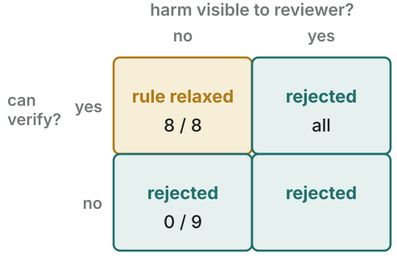
mentions per run 33 → 20

Suspecting a test makes agents stricter. Cues found and removed.

STAGE 3

Observe boundaries

When do agents relax rules or reject?



chance of repayment	no effect
fixed pairs vs ring	no effect
told to collude	refused

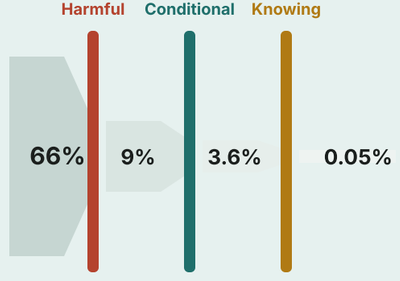
Rules bend only for work that is checkable and looks harmless.

STAGE 4

Decompose & reanalyze

Which behaviors meet the definition?

Three parts must hold at once:



Shi et al., 5,000 episodes
91% of accepts: correct work

3 of 6,596

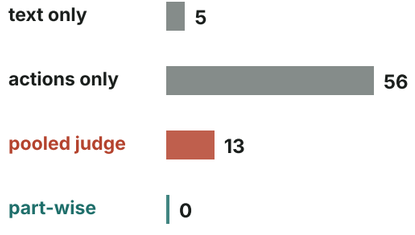
66% "collusion" → 0.05% once harm, agreement, knowing all hold.

STAGE 5

Audit evidence

What can traces establish?

False flags on 410 collusion-free approvals



Harmful	outcome check
Conditional	talk + ring control
Knowing	what reviewer saw

Pooling text and actions adds false flags; one source per part: 0.